

Use Case: Semantic Search Engine for Indian Languages using NLP

Aim:

To develop a semantic search engine that understands user queries in Indian languages and retrieves relevant results based on **meaning (context)** instead of just keywords using NLP techniques.

Code:

```
from sentence_transformers import SentenceTransformer

from sklearn.metrics.pairwise import cosine_similarity

model = SentenceTransformer('all-MiniLM-L6-v2')

documents = [

    "தமிழ் ஒரு பழமையான மொழி",

    "Hindi is widely spoken in India",

    "Machine learning improves search engines"

]

doc_embeddings = model.encode(documents)

query = "Indian languages"

query_embedding = model.encode([query])

scores = cosine_similarity(query_embedding, doc_embeddings)

best_match = documents[scores.argmax()]

print("Best Result:", best_match)
```

Input:

User Query:

"Indian languages"

or

"தமிழ் மொழி"

Output:

Best Result:

"தமிழ் ஒரு பழமையான மொழி"

Relevant journal IEE paper:

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Information retrieval by text summarization for an Indian regional language

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Jagadish S Kallimani ; K G Srinivasa ; Eswara Reddy B All Authors

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Abstract

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Abstract:

The Information Extraction is a method for filtering information from large volumes of text. Information Extraction is a limited task than full text understanding. In full text understanding, we aspire to represent in an explicit fashion about all the information in a text. In contrast, in Information Extraction, we delimit in advance, as part of the specification of the task and the semantic range of the output. In this paper, a model for summarization from large documents using a novel approach has been proposed. Extending the work for an Indian regional language (Kannada) and various analyses of results were discussed.

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